

Wireshark Lab: HTTP and DNS Analysis

Introduction

As described in networking concepts, the **Domain Name System (DNS)** translates hostnames into IP addresses, while the **Hypertext Transfer Protocol (HTTP)** is used for communication between web browsers and web servers.

In real-world web browsing, DNS resolution happens before HTTP communication begins. When a user enters a URL, the system first resolves the domain name using DNS and then establishes a connection to retrieve web content using HTTP.

This lab focuses on analyzing both **DNS and HTTP interactions** using a network analysis approach.

Scenario

A user opens a browser and visits the website:

`http://www.example.com`

The system performs DNS resolution followed by HTTP communication to load the webpage and its associated resources.

Tasks

Perform packet capture using a network analyzer (e.g., Wireshark) and answer the following questions:

Questions

1. Identify the **DNS query message** generated when accessing the website. What is the destination IP address of this query?
2. What is the **destination port number** used for the DNS query?
3. Examine the DNS response. How many **IP addresses (answers)** are returned?
4. Does the DNS response indicate an **authoritative or non-authoritative answer**?
5. After DNS resolution, identify the **HTTP request message** sent by the client. What is the request method used?
6. What is the **requested resource (URL path)** mentioned in the HTTP request?
7. Examine the HTTP request headers. Identify at least **two header fields** and explain their purpose.
8. Analyze the HTTP response message. What is the **status code** returned by the server and what does it indicate?
9. Does the browser send **multiple HTTP requests** to load additional objects (e.g., images, scripts)? Justify your answer from the trace.
10. Explain the **sequence of events** starting from DNS query to HTTP response when loading a webpage.